EX:1.2

AIM:

To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 2 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Drawing setup:

Create a new Fusion 360 design and start a sketch on the XY plane. Set the units to millimetres and use the origin as the centre of the diagram.

2. Outer plate boundary:

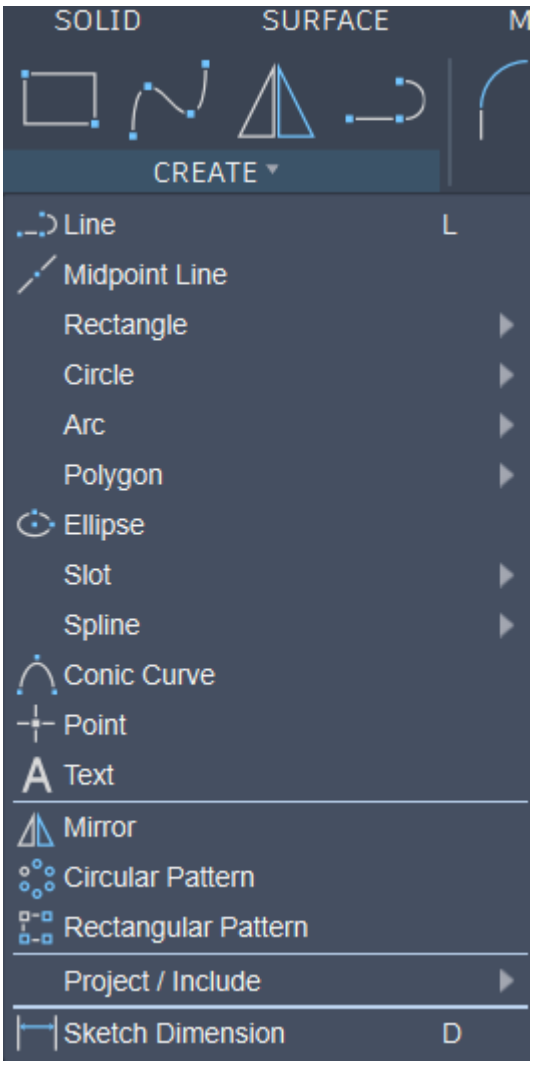
Draw a centred rectangle approximately 178.77 mm wide as a guide for the plate outline. Sketch the sloping sides and rounded top and bottom edges; use tangent constraints where lines meet arcs.

3. Hole placement:

Create four circles with a diameter of 30 mm, matching the four corner holes in the diagram. Position them symmetrically around the centre; use the drawing's dimensions to set their spacing where the extension lines make the reference clear.

4. Centre feature:

Draw the central circle with a diameter of 32 mm at the sketch origin. Add two concentric circles around it to form the surrounding ring, sizing the outer circle to match the diagram.



5. Radial openings:

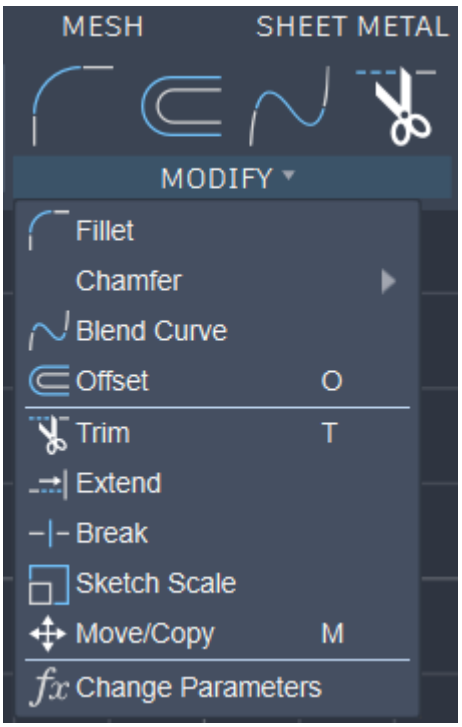
Draw one opening in the ring and use Sketch > Circular Pattern to repeat it around the centre. Set the pattern to six instances, evenly spaced, to match the six openings shown.

6. Sketch constraints:

Apply symmetry about the vertical and horizontal centre lines so corresponding features stay aligned. Use dimensions, tangent constraints, and equal constraints to control the sketch and keep repeated geometry consistent.

7. Finishing the sketch:

Use Trim to remove extra lines and arcs after the outline, holes, and ring openings are in place. Check that the intended plate boundary is closed before selecting Finish Sketch.

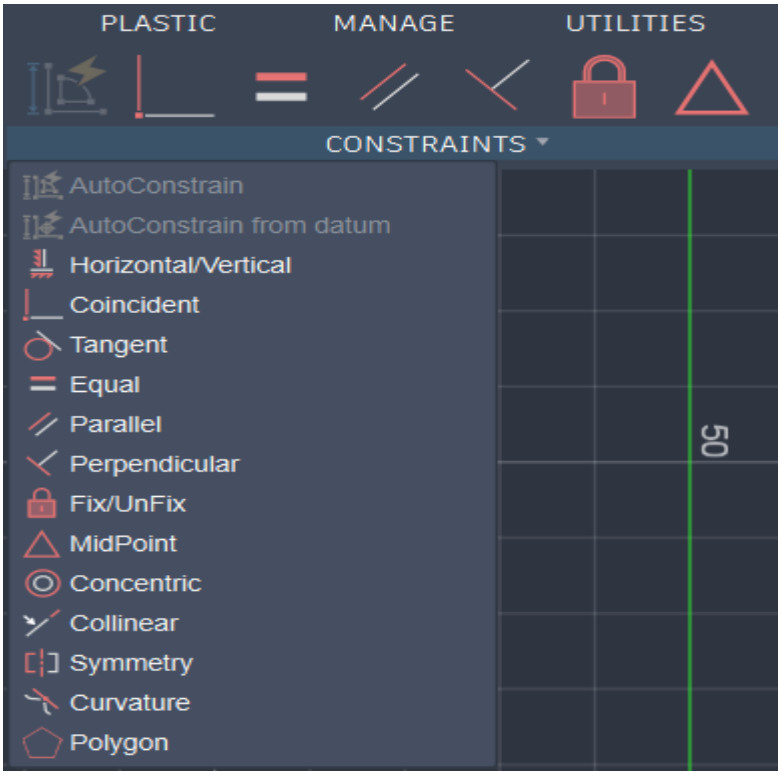


8. Extruding the model:

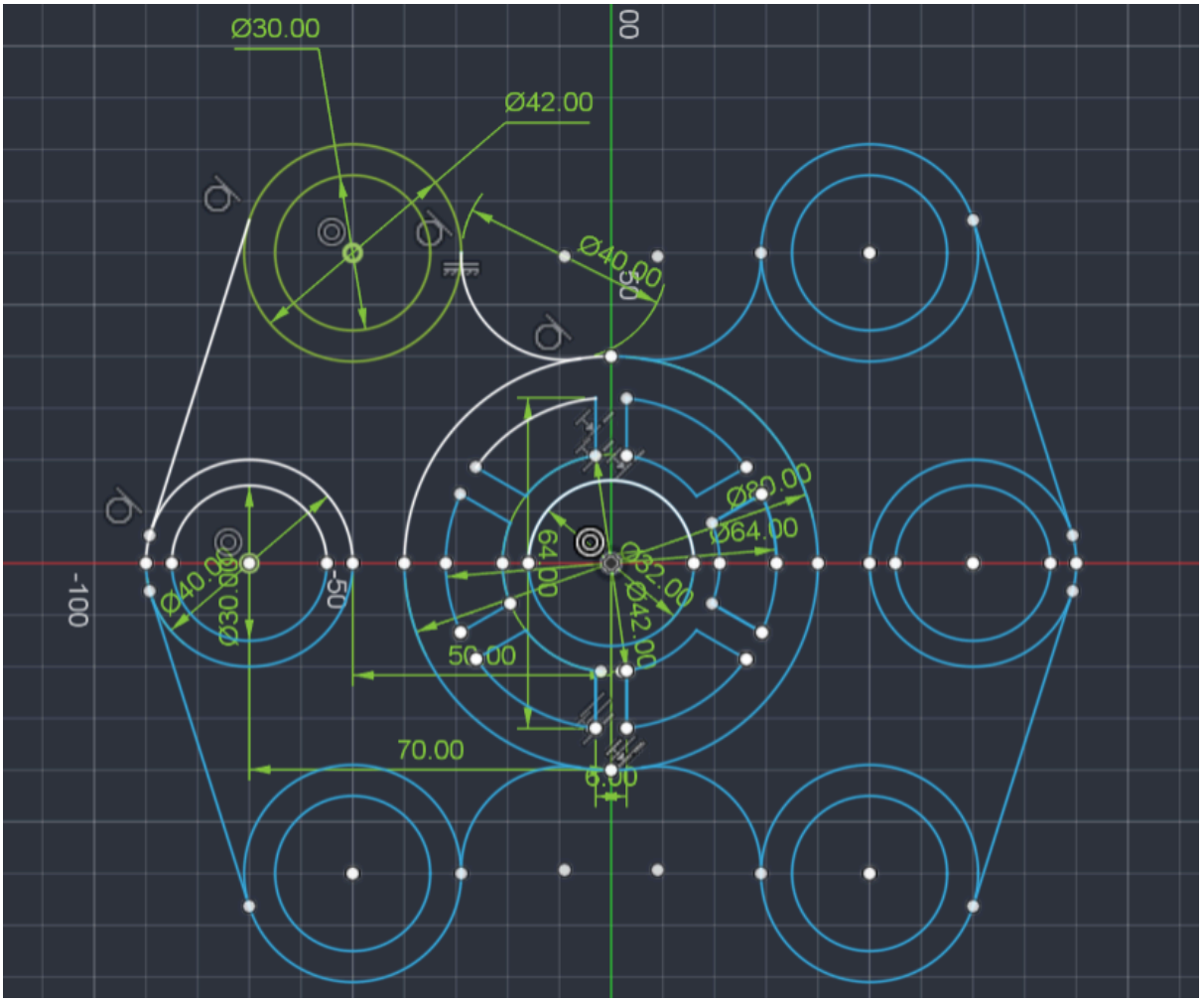
To turn the 2D sketch into a 3D part, select the plate profile and choose Solid > Create > Extrude. Enter the required thickness.

9. Dimension check:

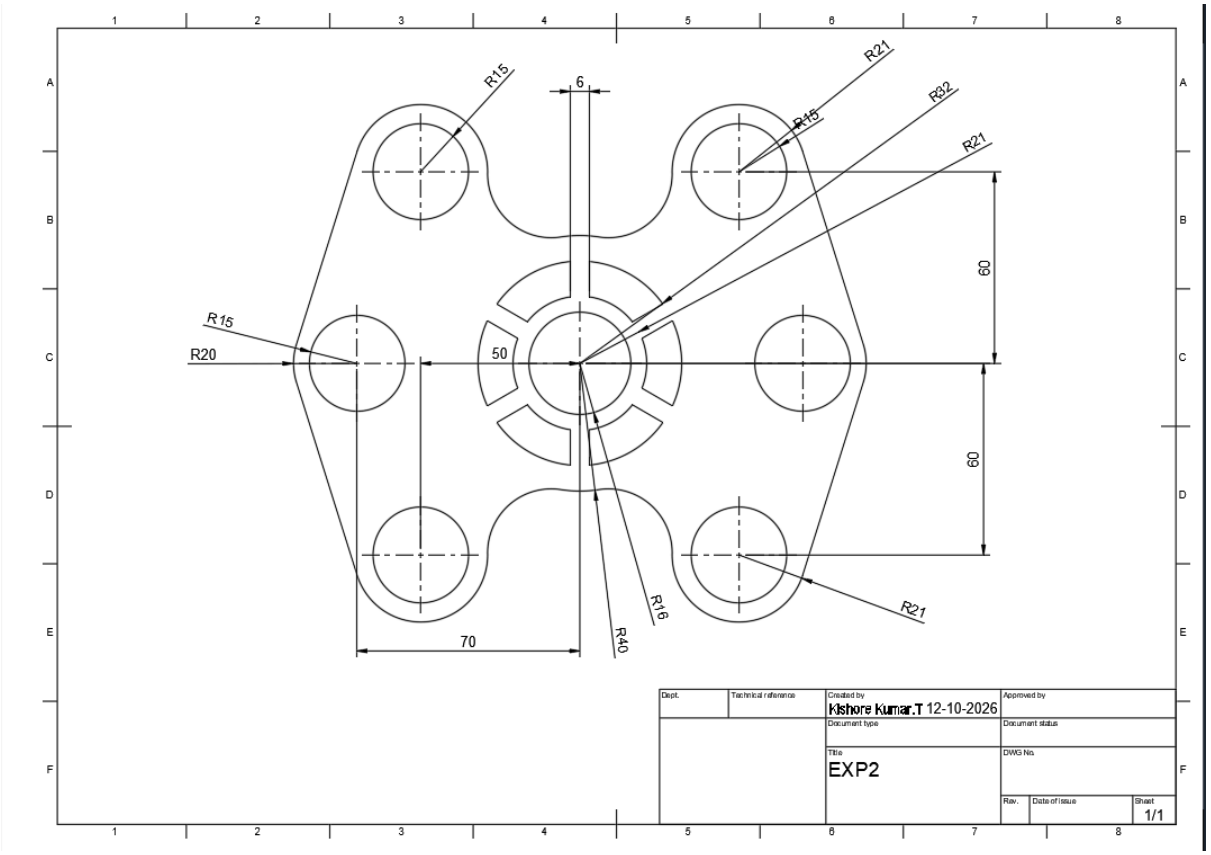
Check the sketch against the dimensions shown, including 178.77 mm, 140 mm, 120 mm, R20, Ø30, and Ø32.



10.Sketch Diagram:



OUTPUT:



RESULT:

Thus the given sketch is drawn and drafted using fusion 360 total.